

Randomized and low-rank approximation

Sharp joint rank and accuracy dependence for strong ℓ_p subspace coresets

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: Challenging because simultaneous control of all subspaces must match joint rank and accuracy lower bounds; community impact is compact robust low-rank fitting.

Partial resolution — 2026-09-13

Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, proves an all-exponent lower bound in [Theorem 1.1](#). For every fixed real $p > 2$, all sufficiently large k and every $0 < \varepsilon < 1/2$, the worst-case minimum support obeys

$$S_p(k, \varepsilon) \geq c_p \frac{k^{p/2}}{\varepsilon^{\beta_p} + (\log k)/k}, \quad \beta_p = \begin{cases} 2, & p \in \{4, 6, 8, \dots\}, \\ 2 - 2/p, & \text{otherwise.} \end{cases}$$

One real input per rank, independent of accuracy, witnesses the bound for arbitrary nonnegative original-row weights. Corollary 1.2 gives a **negative answer to the displayed additive formula for every fixed exponent**, even allowing every fixed logarithmic power. Corollary 1.3 matches the cited upper bound up to logarithms for even $p \geq 4$ when $\varepsilon \geq \sqrt{(\log k)/k}$.

Remaining target: the optimal joint size for non-even exponents and smaller accuracies, including even exponents, is still undetermined. Thus the full classification remains partially resolved. The original statement is retained below.

The [separate independent Codex AI-agent audit](#) passed the proof at this scope. This is informal automated review, not external human peer review or formal verification; no Lean verification was performed. [Proof source](#) · [Attribution, verified affiliation and reproducibility](#).

Original target

Fix a real $p > 2$. For $A \in \mathbb{R}^{n \times d}$, with rows a_i^T , and a linear subspace $F \subseteq \mathbb{R}^d$, define

$$\text{cost}_A(F) = \sum_{i=1}^n \|a_i^T(I - P_F)\|_2^p,$$

where P_F is the Euclidean orthogonal projector. A strong row coreset consists of nonnegative weights $w_1, \dots, w_n$, supported on a small subset of the original rows, for which

$$(1 - \varepsilon) \text{cost}_A(F) \leq \sum_i w_i \|a_i^T(I - P_F)\|_2^p \leq (1 + \varepsilon) \text{cost}_A(F)$$

holds simultaneously for every F of dimension at most k .

Determine the optimal worst-case coreset size as a joint function of k and ε , up to logarithmic factors. In particular, do constants $C_p, c_p > 0$ exist such that, for all n, d , integers $1 \leq k < d$, all input matrices A , and all $0 < \varepsilon < 1/2$, such a coreset exists with

$$|\text{supp}(w)| \leq C_p \left(\frac{k^{p/2}}{\varepsilon} + \frac{k}{\varepsilon^2} \right) \log^{c_p} \left(\frac{2k}{\varepsilon} \right)?$$

The constants may depend on fixed p , but not on n, d, k, ε . This is an existence/size question; no extra running-time target is imposed.

Such a coresets permits subsequent robust low-rank fitting on a much smaller weighted matrix. The August 2026 paper proves an upper bound of order $\tilde{O}_p(k^{p/2}/\varepsilon^2)$ and explicitly identifies the remaining gap to the joint lower-bound expression in Section 1.4. Its new result supersedes older questions that merely asked for an ε^{-2} dependence.

References

1. H. Lin, V. Mirrokni, and D. P. Woodruff, *Nearly Optimal Strong Coresets for ℓ_p Subspace Approximation*, arXiv:2608.26047v2, Section 1.4 “Open problems”; Section 1.2 for the new bounds. [Paper](#).
2. D. P. Woodruff and T. Yasuda, *Root ridge leverage score sampling for ℓ_p subspace approximation*, FOCS 2025, full version arXiv:2407.03262. [Paper](#).

Status check — 2026-09-08

Searched “strong coresets subspace approximation p greater than 2 epsilon 2026” and the exact 2026 title; checked the latest arXiv v2 of August 27, 2026. No later resolution was found. The target comes from the remaining question in that paper, rather than its already improved predecessor bounds.

Audit — 2026-09-10

Rechecked [Lin–Mirrokni–Woodruff, §1.4](#) and the August 27 version record. The joint k, ε gap remains explicit. Strong-coreset follow-up searches found no resolution. Difficulty was raised because the target concerns all input matrices and requires a new uniform sampling analysis.